

Graphs

Dt 14-7-2028

① Cryptographic Processing Time
including 4 curves showing
encryption time, Decryption time,
Signature Generation and signature
Verification time

with ~~ch~~ Varying ~~ch~~ chunk size
(KB $\Rightarrow$ 64, 128, 256,
512, 1024)
(Vs) Processing time

② Authentication Performance $\Rightarrow$ 2 curves
Number of Authentication Requests.
(X-axis) Vs Authentication Delay (left y-axis)
and Authentication Success Rate (%)
(right y-axis)
~~compare~~ proposed method with
~~comp.~~

③ Blockchain Registration Performance
Number of Content chunks Vs $\Rightarrow$ 2 curves
Registration time (left y-axis)
Blockchain transaction (right y-axis)
cost
Comparing Conventional Blockchain Registration
and proposed Registration

④ Blockchain storage overhead $\Rightarrow$ 3 Bar graph
storage methods (full metadata, chunk
hashes, Merkle Root)
Vs. Blockchain storage Size

⑤ Security Overhead Comparison

comparisons $\Rightarrow$ RSA based authentication
ECC based authentication
Existing Blockchain ICM
proposed Blockchain.

Number of content chunks (X-axis) Vs.
Total security overhead (ms or %) (Y-axis)

⑥ Key management overhead

Number of content chunks (100-1000)

Vs. Key management overhead

(One key per chunk,
chunk group key,
One session key per
content)

memory in KB or Number
of managed key

⑦ Security effectiveness

(X-axis) percentage of tampered chunks
(0%, 10%, 20%, 30%, 40%, 50%)

Vs. Tampered chunk detection Rate (%)
(left Y-axis)

Content verification Accuracy (%)
(right Y-axis)